Venkata Lakshmi Parimala Pasupuleti

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Virginia, USA (open to NYC relocation)

PROFESSIONAL SUMMARY

Software Engineer with 8+ years owning production systems for housing operations and building AI on top of them. At Entrata, I built and maintained the software powering eligibility, leasing, rent calculations, and compliance workflows for 100K+ monthly active users -- the same workflows EliseAI is now automating with AI. I've owned end-to-end feature delivery across 120+ Agile sprints, shipped 20+ full-stack modules from design to deployment, and maintained 99.5% uptime with zero compliance violations across 7 years. Now building production agentic AI: LeaseI is an 8-agent LangGraph pipeline for housing compliance operations (93.6% precision / 87.9% F1 on 1,240 test cases), and AgentProbe is an open-source AI evaluation toolkit with CI-integrated regression gates. Strong in Python, Java, TypeScript, React, and production architecture at scale. AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Autonomous Systems for Property Operations | End-to-End Feature Ownership | Scalable Architecture & System Design | Event-Driven Distributed Systems | Agentic AI & Multi-Agent Orchestration | CI/CD & Continuous Integration | Observability & Production Debugging | Python, Java, TypeScript & React

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building agentic AI pipelines and integrating data systems with back-end and client-facing APIs; debugging integration issues across the application stack; turning recurring implementation patterns into reusable product components.

Magizhs Tech Solution
May 2025 -- Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- **Owned new product features end-to-end from design to deployment:** translated multi-system business logic into a LangGraph agentic pipeline, built direct API integrations with client enterprise systems, deployed to production, and served as primary technical consultant during rollout. Shipped in a startup-speed environment with full autonomy over architecture decisions.
- Built event-driven validation layer enforcing business rules before output delivery; implemented RAG pipeline with cited compliance decisions; turned recurring implementation learnings into reusable product components handed off to the core engineering team.

George Mason University
Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; designed operational dashboards (+40% comprehension).

Entrata (Housing Technology SaaS -- 100K+ monthly active users)
Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Built software powering autonomous property operations for 7 years** -- shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for leasing, eligibility, rent calculations, compliance workflows, and document management. Maintained 99.5% uptime. Zero compliance violations across the entire ownership period.
- **Owned HUD income-limit and eligibility pipeline end-to-end:** a few thousand federal rows expanded into ~180,000 domain records powering rent, HAP, and tax-credit calculations for 100K+ housing units. Every release required baseline validation, reconciliation, and rollback preparation -- one wrong import affects thousands of residents' eligibility decisions.
- **Architected scalable, reliable backend systems:** RabbitMQ event-driven async pipelines with idempotency and retry guarantees; Docker microservices redesign (50% scalability gain, 30% deployment time reduction); 200+ SQL query optimizations (30% performance gain); CI/CD pipelines via GitHub Actions; observability tooling that cut mean debug time 40%. Led AWS S3 document migration (100% PHI compliance, zero security incidents).
- Mentored 5+ engineers via 300+ code reviews; built internal debugging tools reducing bug count below 10 for 6+ consecutive months.

PROJECTS

LeaseI: Agentic AI Platform for Housing Compliance Operations
LangGraph 8-agent -- FastAPI -- RAG -- Housing AI -- HITL

Production agentic AI platform for the same domain EliseAI operates in: 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Human Escalation) orchestrating across lease documents, compliance rules, configuration systems, and historical incident data. Core design rule enforced in code: LLM extracts and reasons; deterministic logic or a human makes every final eligibility/compliance call. Results: 93.6% precision / 87.9% F1 on 1,240 test cases across 5 US states; 100% citation correctness. Built end-to-end with no PM, no roadmap -- startup mindset applied to a production AI problem.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: Open-Source AI Evaluation & Observability Toolkit

FastAPI -- LangGraph -- ChromaDB -- CI/CD -- Production Tooling

Production evaluation toolkit for AI agents: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), observability dashboard. Built to measure whether deployed AI achieves its operational objectives -- used to evaluate LeaseI across 1,240+ test cases. Zero cost, fully local.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: NLP-Driven Medical Records Query System

Flask -- LLMs -- BERT -- REST API -- Docker

Built an NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings -- direct integration with clinical records systems, HIPAA-compliant RBAC across multi-system data stores. ~60% reduction in manual retrieval time.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

Languages & Frameworks: Python, Java, TypeScript, JavaScript, React.js, Node.js, PHP, SQL, FastAPI, Flask, Django, Spring Boot, C#

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, LLMs, RAG Pipelines, HITL, Observability, Prompt Engineering, Function Calling, AI Guardrails, FastAPI, MCP, OpenAI API, Anthropic API, HuggingFace, PyTorch

Architecture & Infrastructure: Microservices, Event-Driven Architecture, RabbitMQ, REST APIs, Scalable System Design, Distributed Systems, Continuous Integration, CI/CD (GitHub Actions), Docker, Kubernetes, AWS (S3, EC2, Lambda, RDS, Glue), PostgreSQL, Redis, Linux, Production Debugging

Housing Domain: Eligibility & Rent Calculation Workflows, HUD Compliance, LIHTC / Tax Credit, Affordable Housing Operations, Property Management Systems, Leasing Workflows, HAP Calculations, Tenant Data Pipelines